Unit 2 End-of-Unit Assessment - Form A Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: A learner records the length of every pencil in a box. Is asking about these lengths statistical? Name the population, variable, and unit.

Solution / evidence: Yes; anticipated variability across lengths. Population: pencils in the box. Variable: length. Unit: centimeters or another appropriate length unit.

Item 2

Prompt: Group 4, 10, 12, 20, 21, 28 into 0–9, 10–19, 20–29. Give frequencies and explain the interval boundary at 10.

Solution / evidence: 1,2,3. Ten belongs only in 10–19. This form uses a value below 10 for the first observation.

Item 3

Prompt: Find the median of 10, 4, 8, 6. Show the ordering and explain the two middle positions.

Solution / evidence: Order 4, 6, 8, 10; median 7.

Item 4

Prompt: Find the five-number summary for 4, 6, 8, 10, 12, 14, 16. Exclude the overall median from both halves.

Solution / evidence: 4, 6, 10, 14, 16.

Item 5

Prompt: Compare IQR for [4, 6, 8, 10] and [4, 6, 8, 30]. Use medians of the two halves. Explain whether changing the maximum can affect IQR.

Solution / evidence: First IQR 4; second IQR 14. It can affect Q3 in this four-value set.

Item 6

Prompt: Three observations are 4, 6, 8. Find a fourth so the final mean is 7. Check it.

Solution / evidence: Target total 28; known total 18; fourth 10.

Item 7

Prompt: Find and interpret the MAD of 4, 6, 8, 10.

Solution / evidence: Mean 7; distances 3,1,1,3; MAD 2 units, average distance from the mean.

Item 8

Prompt: A: 4, 5, 6, 7, 28. Find mean and median. Choose a center for a central observation and explain a purpose for the other measure.

Solution / evidence: Mean 10, median 6. Median describes a central observation with less influence from the extreme. Mean remains useful for equal sharing or total divided by count.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.